Packet Tracer - Propagate a Default Route in OSPFv2

Addressing Table

Device	Interface	IPv4 Address	Subnet Mask	Default Gateway
R1	G0/0	172.16.1.1	255.255.255.0	N/A
	S0/0/0	172.16.3.1	255.255.255.252	
	S0/0/1	192.168.10.5	255.255.255.252	
R2	G0/0	172.16.2.1	255.255.255.0	N/A
	S0/0/0	172.16.3.2	255.255.255.252	
	S0/0/1	192.168.10.9	255.255.255.252	
	S0/1/0	209.165.200.225	255.255.255.224	
R3	G0/0	192.168.1.1	255.255.255.0	N/A
	S0/0/0	192.168.10.6	255.255.255.252	
	S0/0/1	192.168.10.10	255.255.255.252	
PC1	NIC	172.16.1.2	255.255.255.0	172.16.1.1
PC2	NIC	172.16.2.2	255.255.255.0	172.16.2.1
PC3	NIC	192.168.1.2	255.255.255.0	192.168.1.1
Web Server	NIC	64.100.1.2	255.255.255.0	64.100.1.1

Objectives

Part 1: Propagate a Default Route

Part 2: Verify Connectivity

Background

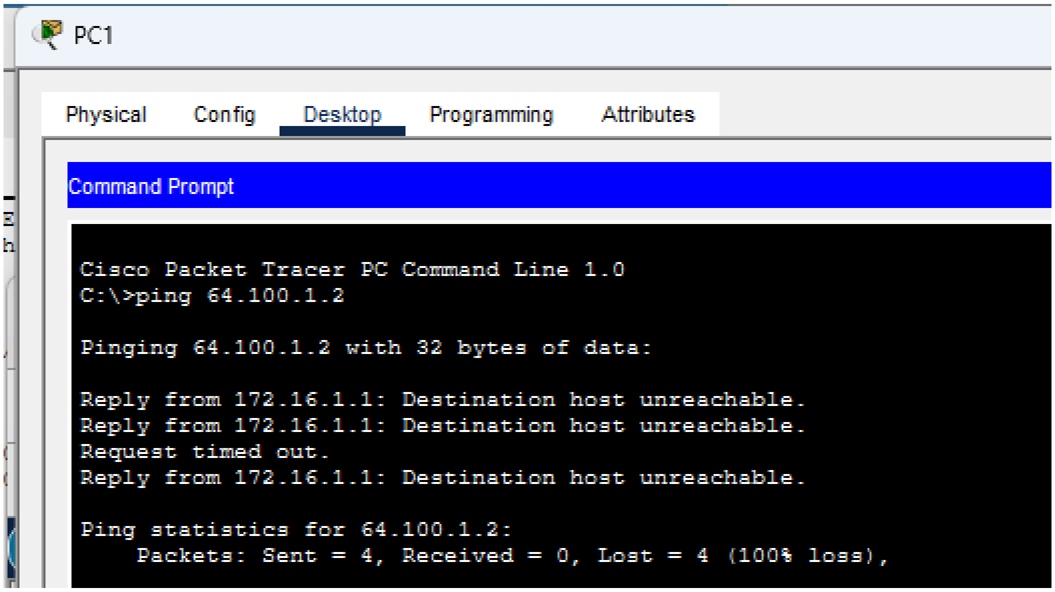
In this activity, you will configure an IPv4 default route to the Internet and propagate that default route to other OSPF routers. You will then verify the default route is in downstream routing tables and that hosts can now access a web server on the Internet.

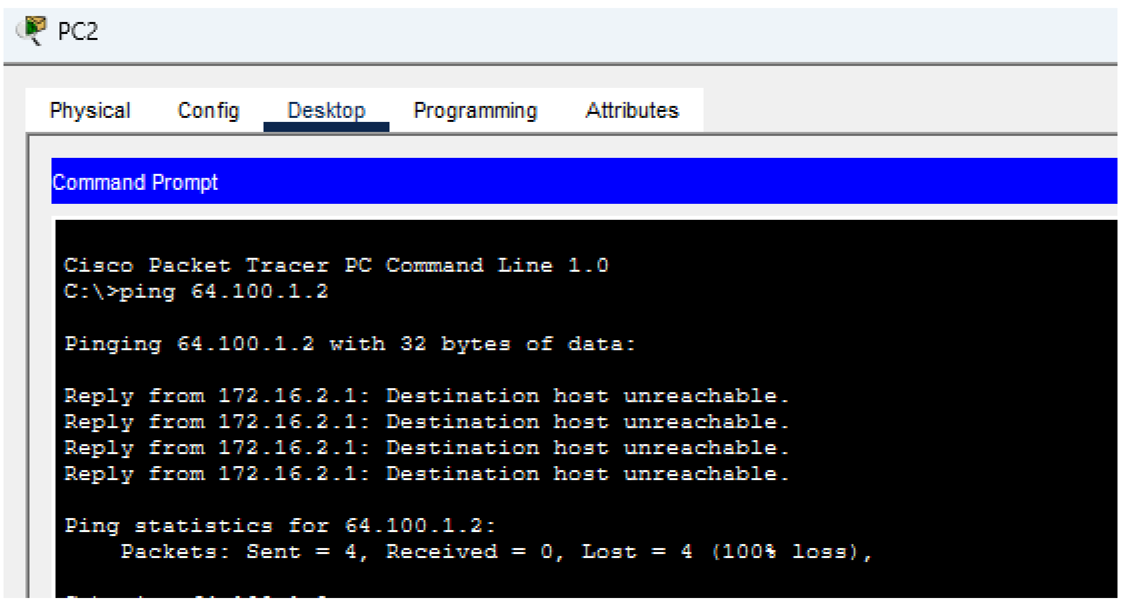
Instructions

Part 1: Propagate a Default Route

Step 1: Test connectivity to the Web Server

a. From PC1, PC2, and PC3, attempt to ping the Web Server IP address, 64.100.1.2.


PC1
Physical
Config
Desktop
Programming
Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 64.100.1.2
Pinging 64.100.1.2 with 32 bytes of data:
Reply from 172.16.1.1: Destination host unreachable.
Reply from 172.16.1.1: Destination host unreachable.
Request timed out.
Reply from 172.16.1.1: Destination host unreachable.
Ping statistics for 64.100.1.2:
Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),


PC2
Physical
Config
Desktop
Programming
Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 64.100.1.2
Pinging 64.100.1.2 with 32 bytes of data:
Reply from 172.16.2.1: Destination host unreachable.
Reply from 172.16.2.1: Destination host unreachable.
Reply from 172.16.2.1: Destination host unreachable.
Reply from 172.16.2.1: Destination host unreachable.
Ping statistics for 64.100.1.2:
Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

PC3

Physical Config Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 64.100.1.2

Pinging 64.100.1.2 with 32 bytes of data:

Reply from 192.168.1.1: Destination host unreachable.
Reply from 192.168.1.1: Destination host unreachable.
Reply from 192.168.1.1: Destination host unreachable.
Reply from 192.168.1.1: Destination host unreachable.

Ping statistics for 64.100.1.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 64.100.1.2
```

Were any of the pings successful?

What message did you receive, and which device issued the message?

b. Examine the routing tables on routers R1, R2, and R3.

```
R1>
R1>enable
R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     172.16.0.0/16 is variably subnetted, 5 subnets, 3 masks
C       172.16.1.0/24 is directly connected, GigabitEthernet0/0
L       172.16.1.1/32 is directly connected, GigabitEthernet0/0
O       172.16.2.0/24 [110/65] via 172.16.3.2, 00:06:43, Serial0/0/0
C       172.16.3.0/30 is directly connected, Serial0/0/0
L       172.16.3.1/32 is directly connected, Serial0/0/0
O    192.168.1.0/24 [110/65] via 192.168.10.6, 00:06:43, Serial0/0/1
     192.168.10.0/24 is variably subnetted, 3 subnets, 2 masks
C       192.168.10.4/30 is directly connected, Serial0/0/1
L       192.168.10.5/32 is directly connected, Serial0/0/1
O       192.168.10.8/30 [110/128] via 192.168.10.6, 00:06:43, Serial0/0/1
                        [110/128] via 172.16.3.2, 00:06:43, Serial0/0/0
     209.165.200.0/27 is subnetted, 1 subnets
O       209.165.200.224/27 [110/128] via 172.16.3.2, 00:06:43, Serial0/0/0
```

```
R2>enable
R2#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     172.16.0.0/16 is variably subnetted, 5 subnets, 3 masks
O       172.16.1.0/24 [110/65] via 172.16.3.1, 00:07:26, Serial0/0/0
C       172.16.2.0/24 is directly connected, GigabitEthernet0/0
L       172.16.2.1/32 is directly connected, GigabitEthernet0/0
C       172.16.3.0/30 is directly connected, Serial0/0/0
L       172.16.3.2/32 is directly connected, Serial0/0/0
O    192.168.1.0/24 [110/65] via 192.168.10.10, 00:07:26, Serial0/0/1
     192.168.10.0/24 is variably subnetted, 3 subnets, 2 masks
O       192.168.10.4/30 [110/128] via 172.16.3.1, 00:07:26, Serial0/0/0
                        [110/128] via 192.168.10.10, 00:07:26, Serial0/0/1
C       192.168.10.8/30 is directly connected, Serial0/0/1
L       192.168.10.9/32 is directly connected, Serial0/0/1
     209.165.200.0/24 is variably subnetted, 2 subnets, 2 masks
C       209.165.200.224/27 is directly connected, Serial0/1/0
L       209.165.200.225/32 is directly connected, Serial0/1/0

R2#configure terminal
```

```
R3#
R3#R2>enable
Translating "R2>enable"...domain server (255.255.255.255)
% Unknown command or computer name, or unable to find computer address

R3#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 192.168.10.9 to network 0.0.0.0

     172.16.0.0/16 is variably subnetted, 3 subnets, 2 masks
O       172.16.1.0/24 [110/65] via 192.168.10.5, 00:19:17, Serial0/0/0
O       172.16.2.0/24 [110/65] via 192.168.10.9, 00:19:17, Serial0/0/1
O       172.16.3.0/30 [110/128] via 192.168.10.9, 00:19:17, Serial0/0/1
                      [110/128] via 192.168.10.5, 00:19:17, Serial0/0/0
     192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.1.0/24 is directly connected, GigabitEthernet0/0
L       192.168.1.1/32 is directly connected, GigabitEthernet0/0
     192.168.10.0/24 is variably subnetted, 4 subnets, 2 masks
C       192.168.10.4/30 is directly connected, Serial0/0/0
L       192.168.10.6/32 is directly connected, Serial0/0/0
C       192.168.10.8/30 is directly connected, Serial0/0/1
L       192.168.10.10/32 is directly connected, Serial0/0/1
     209.165.200.0/27 is subnetted, 1 subnets
O       209.165.200.224/27 [110/128] via 192.168.10.9, 00:19:17, Serial0/0/1
```

What statement is present in the routing tables that indicates that the pings to the Web Server will fail?

Step 2: Configure a default route on R2.

Configure **R2** with a directly attached default route to the Internet.

```
R2(config)# ip route 0.0.0.0 0.0.0.0 Serial0/1/0
```

```
R2#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#ip route 0.0.0.0 0.0.0.0 Serial0/1/0
%Default route without gateway, if not a point-to-point interface, may impact performance
```

Note: Router will give a warning that if this interface is not a point-to-point connection, it may impact performance. You can ignore this warning because it is a point-to-point connection.

Step 3: Propagate the route in OSPF.

Configure OSPF to propagate the default route in OSPF routing updates.

```
R2(config)# router ospf 1
R2(config-router)# default-information originate
```

```
%Default route without gateway, if not a point-to-point interface, may imp
R2(config)#router ospf 1
R2(config-router)#default-information originate
R2(config-router)#end
R2#
%SYS-5-CONFIG_I: Configured from console by console

R2#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
R2#
```

Step 4: Examine the routing tables on R1 and R3.

Examine the routing tables of **R1** and **R3** to verify that the route has been propagated.

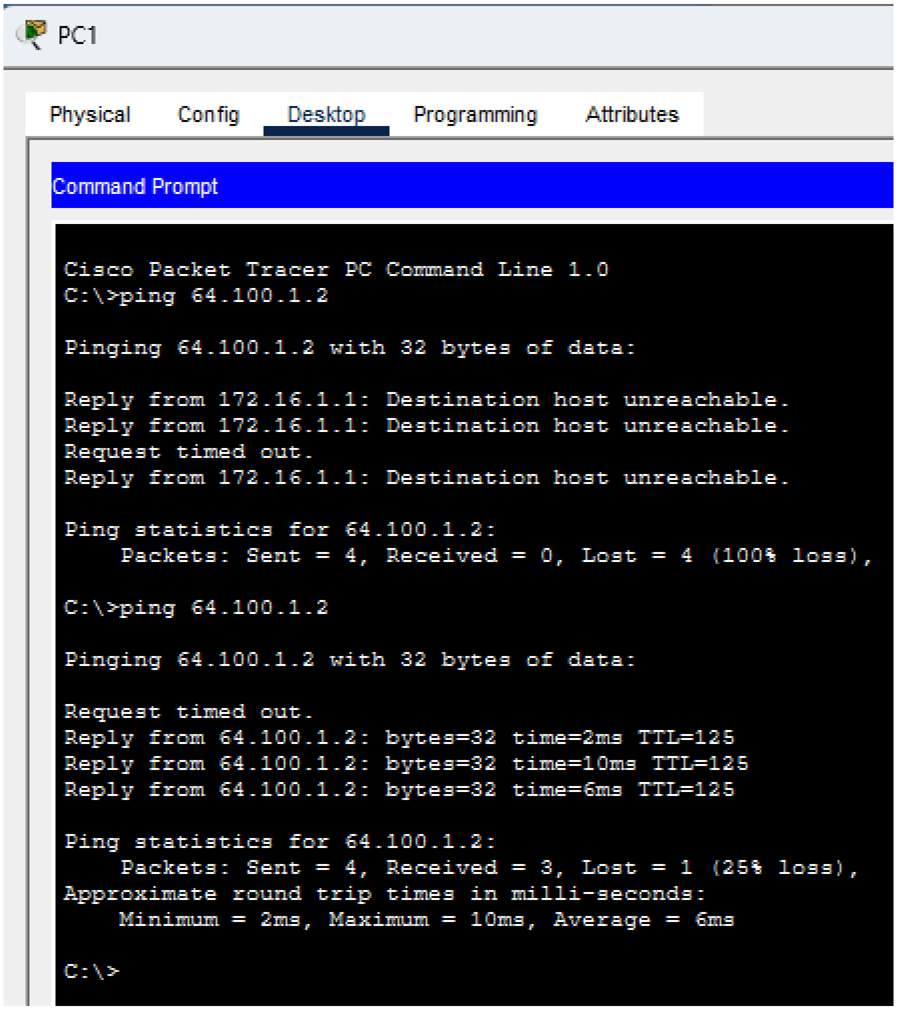
```
R1> show ip route
<output omitted>
Gateway of last resort is 172.16.3.2 to network 0.0.0.0
<output omitted>
O*E2 0.0.0.0/0 [110/1] via 172.16.3.2, 00:00:08, Serial0/0/0
!-------------------
R3> show ip route
<output omitted>
Gateway of last resort is 192.168.10.9 to network 0.0.0.0
<output omitted>
O*E2 0.0.0.0/0 [110/1] via 192.168.10.9, 00:08:15, Serial0/0/1
```

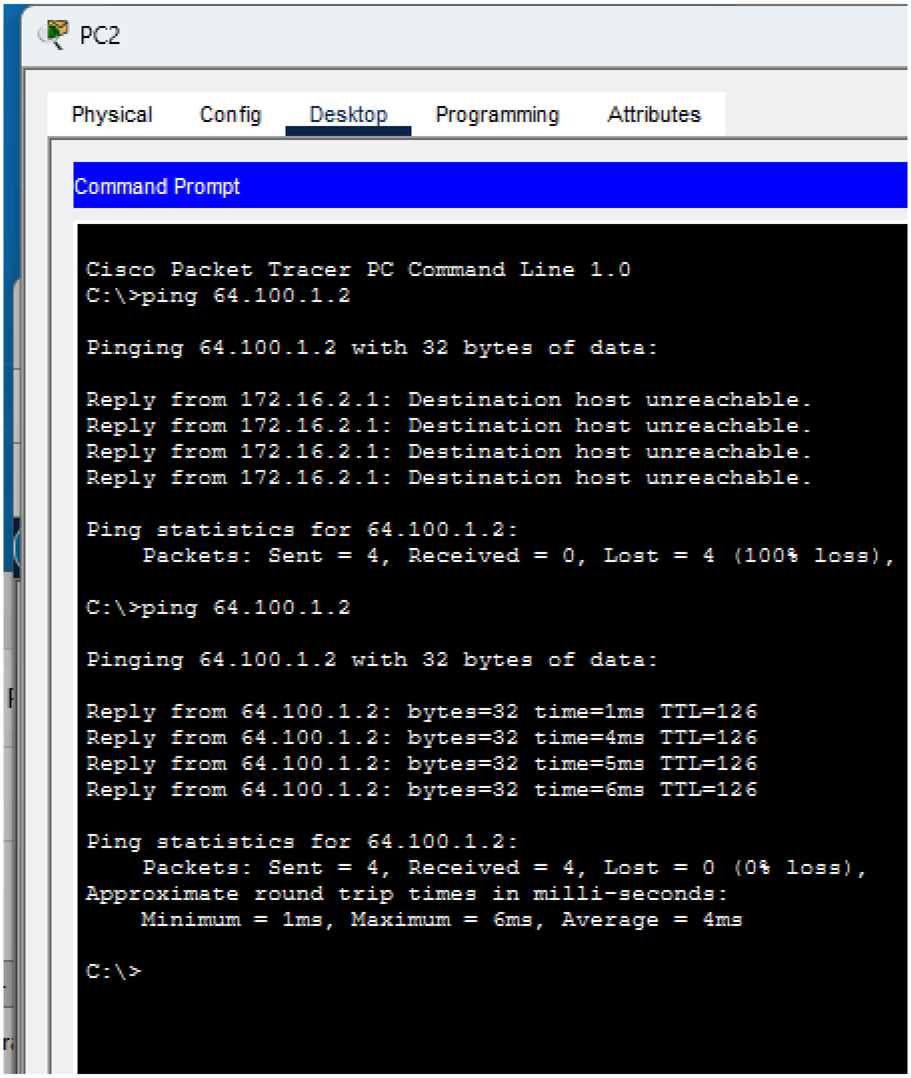
```
R3#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 192.168.10.9 to network 0.0.0.0

     172.16.0.0/16 is variably subnetted, 3 subnets, 2 masks
O       172.16.1.0/24 [110/65] via 192.168.10.5, 00:19:17, Serial0/0/0
O       172.16.2.0/24 [110/65] via 192.168.10.9, 00:19:17, Serial0/0/1
O       172.16.3.0/30 [110/128] via 192.168.10.9, 00:19:17, Serial0/0/1
                      [110/128] via 192.168.10.5, 00:19:17, Serial0/0/0
     192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.1.0/24 is directly connected, GigabitEthernet0/0
L       192.168.1.1/32 is directly connected, GigabitEthernet0/0
     192.168.10.0/24 is variably subnetted, 4 subnets, 2 masks
C       192.168.10.4/30 is directly connected, Serial0/0/0
L       192.168.10.6/32 is directly connected, Serial0/0/0
C       192.168.10.8/30 is directly connected, Serial0/0/1
L       192.168.10.10/32 is directly connected, Serial0/0/1
     209.165.200.0/27 is subnetted, 1 subnets
O       209.165.200.224/27 [110/128] via 192.168.10.9, 00:19:17, Serial0/0/1
 --More--
```

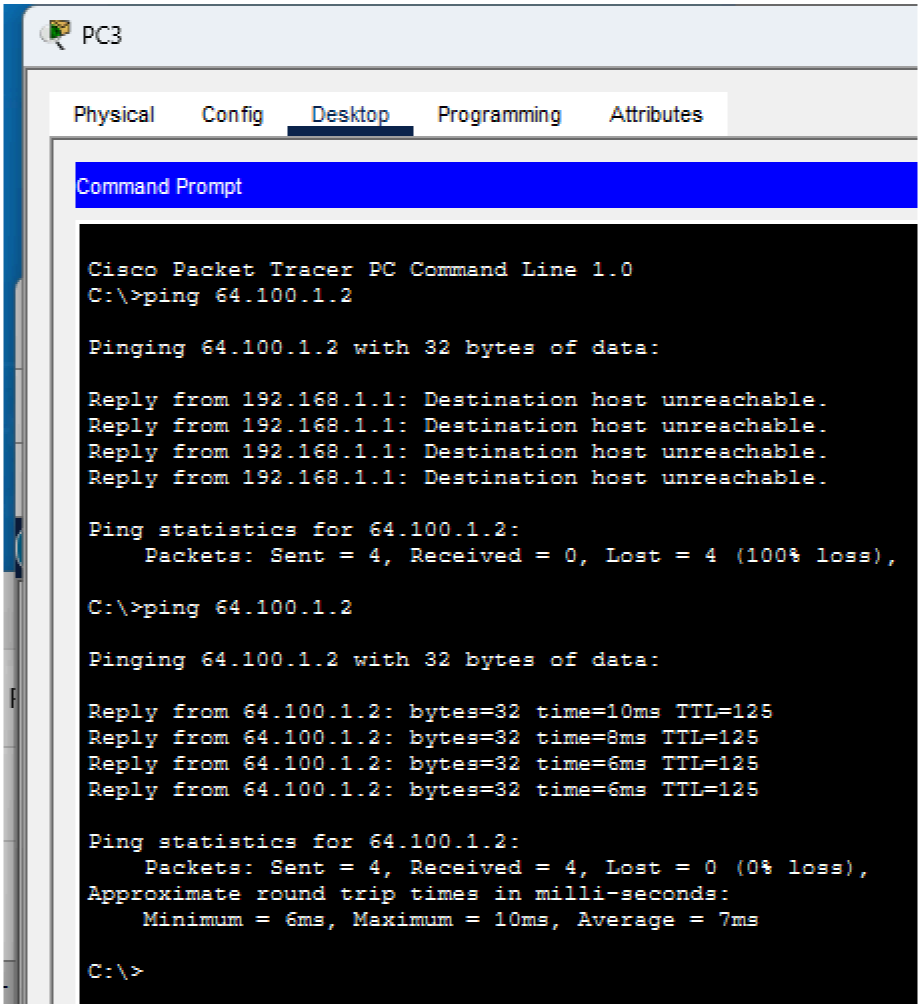
```
R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 172.16.3.2 to network 0.0.0.0

     172.16.0.0/16 is variably subnetted, 5 subnets, 3 masks
C       172.16.1.0/24 is directly connected, GigabitEthernet0/0
L       172.16.1.1/32 is directly connected, GigabitEthernet0/0
O       172.16.2.0/24 [110/65] via 172.16.3.2, 00:11:01, Serial0/0/0
C       172.16.3.0/30 is directly connected, Serial0/0/0
L       172.16.3.1/32 is directly connected, Serial0/0/0
O    192.168.1.0/24 [110/65] via 192.168.10.6, 00:11:01, Serial0/0/1
     192.168.10.0/24 is variably subnetted, 3 subnets, 2 masks
C       192.168.10.4/30 is directly connected, Serial0/0/1
L       192.168.10.5/32 is directly connected, Serial0/0/1
O       192.168.10.8/30 [110/128] via 192.168.10.6, 00:11:01, Serial0/0/1
                        [110/128] via 172.16.3.2, 00:11:01, Serial0/0/0
     209.165.200.0/27 is subnetted, 1 subnets
O       209.165.200.224/27 [110/128] via 172.16.3.2, 00:11:01, Serial0/0/0
O*E2 0.0.0.0/0 [110/1] via 172.16.3.2, 00:01:51, Serial0/0/0

R1#
```

Part 2: Verify Connectivity

Verify that **PC1**, **PC2**, and **PC3** can ping the web server.


PC1
Physical
Config
Desktop
Programming
Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 64.100.1.2
Pinging 64.100.1.2 with 32 bytes of data:
Reply from 172.16.1.1: Destination host unreachable.
Reply from 172.16.1.1: Destination host unreachable.
Request timed out.
Reply from 172.16.1.1: Destination host unreachable.
Ping statistics for 64.100.1.2:
Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
C:\>ping 64.100.1.2
Pinging 64.100.1.2 with 32 bytes of data:
Request timed out.
Reply from 64.100.1.2: bytes=32 time=2ms TTL=125
Reply from 64.100.1.2: bytes=32 time=10ms TTL=125
Reply from 64.100.1.2: bytes=32 time=6ms TTL=125
Ping statistics for 64.100.1.2:
Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
Minimum = 2ms, Maximum = 10ms, Average = 6ms
C:\>


PC2
Physical
Config
Desktop
Programming
Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 64.100.1.2
Pinging 64.100.1.2 with 32 bytes of data:
Reply from 172.16.2.1: Destination host unreachable.
Reply from 172.16.2.1: Destination host unreachable.
Reply from 172.16.2.1: Destination host unreachable.
Reply from 172.16.2.1: Destination host unreachable.
Ping statistics for 64.100.1.2:
Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
C:\>ping 64.100.1.2
Pinging 64.100.1.2 with 32 bytes of data:
Reply from 64.100.1.2: bytes=32 time=1ms TTL=126
Reply from 64.100.1.2: bytes=32 time=4ms TTL=126
Reply from 64.100.1.2: bytes=32 time=5ms TTL=126
Reply from 64.100.1.2: bytes=32 time=6ms TTL=126
Ping statistics for 64.100.1.2:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 1ms, Maximum = 6ms, Average = 4ms
C:\>


PC3
Physical
Config
Desktop
Programming
Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 64.100.1.2
Pinging 64.100.1.2 with 32 bytes of data:
Reply from 192.168.1.1: Destination host unreachable.
Reply from 192.168.1.1: Destination host unreachable.
Reply from 192.168.1.1: Destination host unreachable.
Reply from 192.168.1.1: Destination host unreachable.
Ping statistics for 64.100.1.2:
Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
C:\>ping 64.100.1.2
Pinging 64.100.1.2 with 32 bytes of data:
Reply from 64.100.1.2: bytes=32 time=10ms TTL=125
Reply from 64.100.1.2: bytes=32 time=8ms TTL=125
Reply from 64.100.1.2: bytes=32 time=6ms TTL=125
Reply from 64.100.1.2: bytes=32 time=6ms TTL=125
Ping statistics for 64.100.1.2:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 6ms, Maximum = 10ms, Average = 7ms
C:\>